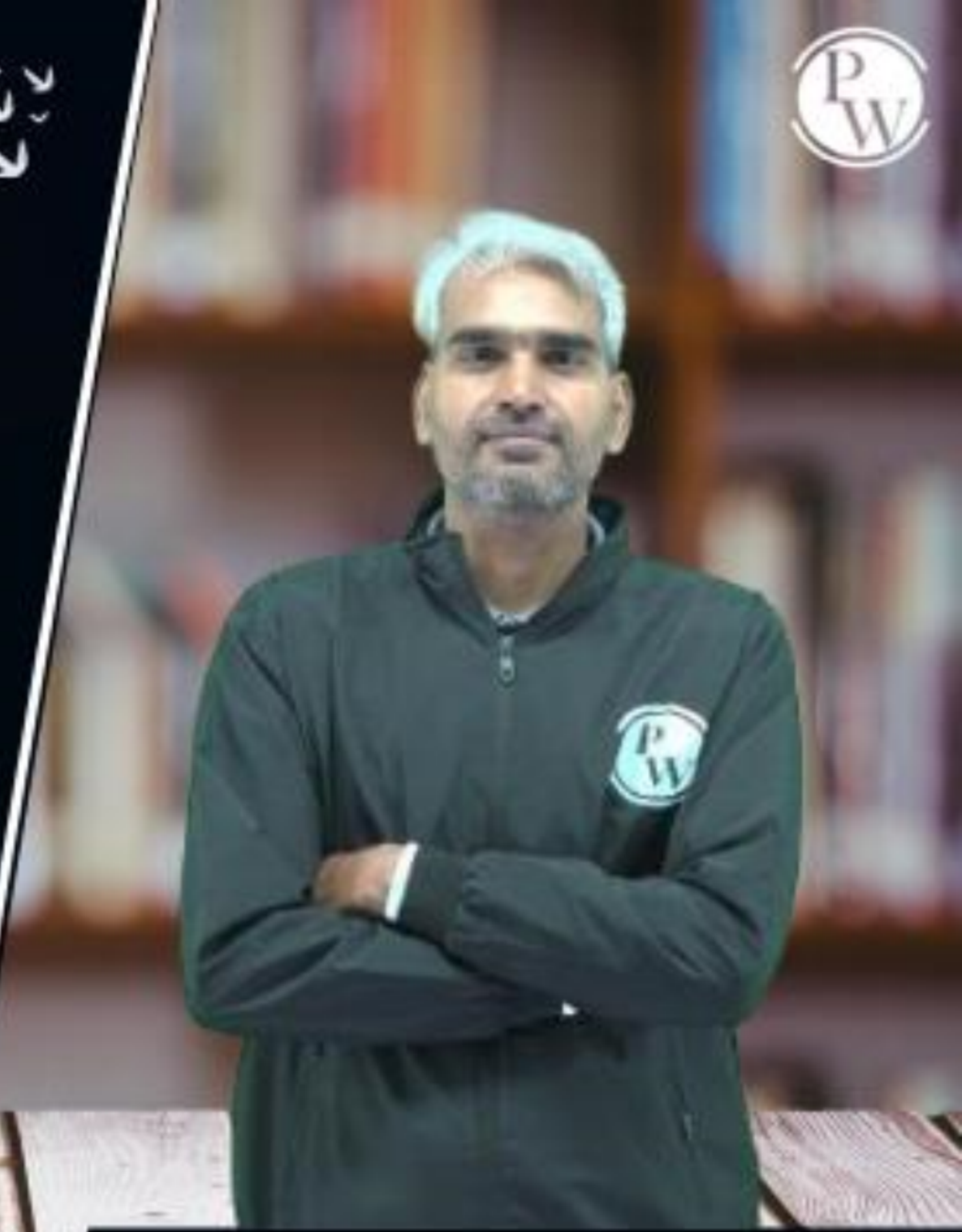
Computer Science & IT

Database Management System

Relational Model & Normal Forms

Lecture No. 03



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Recap of Previous Lecture

- Topic — Relational Database Model
- Topic — Functional dependency
- Topic — Types of functional dependency

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Topics to be Covered

- **Topic** Properties of functional dependency
- **Topic** Different types of keys in RDBMS
- **Topic** Candidate key
- **Topic** Super key

Slide

Topic : Functional dependency (FD)

Functional dependencies are used to specify the relationship among the attributes of a given relation

Functional dependencies are denoted using the format

$$X \longrightarrow Y$$

where X & Y are some set of attributes from the given relation

If $X \longrightarrow Y$ exist in relation R

then "X" is called determinant attribute set

& "y" is called dependent attribute set

Topic : Functional dependency (FD)

- Functional dependency defines the relationship between two sets of attributes in a relational table.

- It states that the values of dependent attribute set can be completely determined by the values of determinant attributes set. In other words, if a determinant attribute's value is known, the value of the dependent attribute can be deduced.

Topic : Functional dependency (FD)

- Let R be the relational schema with X and Y as the attribute sets over relation R.

Functional dependency $X \rightarrow Y$ exists in R only if
For all pair of tuples $t1, t2 \in R$
If $t1.X = t2.X$ then $t1.Y = t2.Y$

IF functional dependency $X \rightarrow Y$ exists in R
then (for all pair of tuples $t_1, t_2 \in R$
if $t_1 \cdot X = t_2 \cdot X$ then $t_1 \cdot Y = t_2 \cdot Y$)

P only if Q ≡ if P then Q

P	Q	$(P \rightarrow Q)$
T	T	T
F	T	T
F	F	T
T	F	F

If $P \to Q$ is true

then is it necessary for $Q \to P$ to be true

Ans is <u>NO</u>.

∴ If, for all pair of tuples $t_1, t_2 \in R$

(if $t_1.X = t_2.X$ then $t_1.Y = t_2.Y$)

then functional dependency $X \to Y$ exists in R

Need Not be true

Topic : Functional dependency (FD)

- If functional dependency $X \rightarrow Y$ exists in the relation then,

  Whenever values w.r.t. attribute set X are same in any pair of tuples, then in those tuples values of Y will also be same

it is the necessary Cond^n for $X \rightarrow Y$ to exist in a relation, but not sufficient

But if values of "Y" are same in two tuples, then values of X need not be same in those tuples

→ Consider the following relation :-

Sid	Sname	Branch
1	Ram	CS
2	Ram	CS
3	John	IT
4	Mohan	CS
5	Sai	IT

Student Relation

If we know value of "Sid" then we can uniquely identify the "Sname" w.r.t. that "Sid"

∴ Functional dependendency

$Sid \rightarrow Sname$

May exist in the relation

Necessary Condⁿ holds true in this relation w.r.t. FD

$\underset{X}{Sid} \longrightarrow \underset{Y}{Sname}$

∴ $Sid \rightarrow Sname$ may hold in the relation

→ Consider the following relation :-

Sid	Sname	Branch
1	Ram	CS
2	Ram	CS
3	John	IT
4	Mohan	CS
5	Sai	IT
6	Ram	IT

Student Relation

In this specific relational instance, If we know the "Sname" then we can uniquely identify "Branch" w.r.t given Sname

∴ Sname → Branch may hold true

Necessary Cond^n holds true in this relation w.r.t. FD

Sname → Branch

∴ Sname → Branch may hold in the relation

→ Consider the following relation:-

Sid	Sname	Branch
1	Ram	CS
2	Ram	CS
3	John	IT
4	Mohan	CS
5	Sai	IT

Student Relation

In this specific relational instance, If we know the "Branch" of the student then We may not uniquely identify the "Sname" w.r.t. given "Branch"

∴ $\boxed{\text{Branch} \rightarrow \text{Sname does not hold (exists) in the given relation}}$

Necessary Condⁿ does not hold true in this relation w.r.t. FD

Branch → Sname

∴ FD, Branch → Sname Can never exist in this relation

Topic : Functional dependency (FD)

- If necessary condition for functional dependency "X→Y" does not hold true based on given relation instance, then functional dependency "X→Y" can never exist in the given relation.

- Even if necessary condition for functional dependency "X→Y" does hold true based on given relation instance, then also we can not be sure whether functional dependency X→Y exists in the relation or not, because it is just the relational instance.

#Q. Consider the following relational instance

	A	B	C
t_1	1	2	3
t_2	1	2	4
	2	2	1
	3	1	2
	4	1	2

$A \not\rightarrow C$

$A \rightarrow B$ ✓

$AB \not\rightarrow C$

$BC \not\rightarrow A$

Which of the following functional dependency may hold true (not necessarily) based on given relational instance.

a) A→ C

b) A → B ✓

c) AB → C

d) BC → A

#Q. Consider the following relational instance

No two values of 'C' are same ∴ we don't need to worry about others

A	B	C
1	1	1
1	2	2
2	4	3
3	3	4
4	1	5
5	3	6

Which of the following functional dependency may hold true (not necessarily) based on given relational instance.

a) $B \rightarrow C$

b) $A \rightarrow B$

c) $C \rightarrow B$

d) $B \rightarrow A$

e) $C \rightarrow A$

#Q. From the following instance of a relation schema R (A,B,C), we can conclude that:

we can conclude, that $B \to C$ can never hold in this relation

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

Necessary Cond^n is true w.r.t. $A \to B$

$\therefore A \to B$ may hold, but we can not conclude that $A \to B$ holds in the relation

(A) A functionally determines B, and B functionally determines C

(B) A functionally determines B, and B does not functionally determine C

(C) B does not functionally determine C

(D) A does not functionally determine B, and B does not functionally determine C

Note: Functional dependencies that holds true in a relation are identified by database designer, and mostly they are given in the question

Topic : Types of Functional Dependency

* There are two types of functional dependencies

1. Trivial functional dependency
2. Non-trivial functional dependency

We can define a third type of functional dependency which is actually a mixture of trivial & non-trivial.

∴ We can call it "Semi-non-trivial functional dependency"

Topic : Trivial Functional Dependency

If X and Y are two sets of attributes over relation R such that $X \supseteq Y$,

then $X \longrightarrow Y$ is called a trivial FD.

Every trivial FD possible from the attributes of a relation will always hold true in that relation but they will not provide any useful information

eg: Sid, Sname $\longrightarrow$ Sid

Sid, Sname $\longrightarrow$ Sname

Sid, Sname $\longrightarrow$ Sid, Sname

Topic : Non-trivial Functional Dependency

FD $X \rightarrow Y$ is called a non-trivial FD in relation R

Only if, (i) X and Y are non-Empty set of attributes over relation R

and (ii) $X \cap Y = \phi$

eg. R(Sid, Sname, Branch) ⇒ Possible non-trivial FDs w.r.t. attributes of relⁿ R are

Sid → Sname, Sid → Branch, Sname → Sid | Sname → Branch, Branch → Sid, Branch → Sname | Sid → Sname, Branch, Sname → Sid, Branch, Branch → Sid, Sname | Sid, Sname → Branch, Sid, Branch → Sname, Sname, Branch → Sid

Every non-trivial FD need not hold true in the given relation

Topic : Semi non-trivial Functional Dependency

FD $X \rightarrow Y$ can be called a semi non-trivial FD

if and only if

(i) $X \not\supseteq Y$

and (ii) $X \cap Y \neq \phi$

eg. R(Sid, Sname, Branch)

Sid, Sname $\longrightarrow$ Sname, Branch

$\{Sid, Sname\} \not\supseteq \{Sname, Branch\}$

$\{Sid, Sname\} \cap \{Sname, Branch\} = \{Sname\} \neq \phi$

Both condⁿ are satisfied

∴ we can call it semi non-trivial FD.

#Q. Consider the following relational instance

$X \to Y$

Possible non-trivial FDs w.r.t attributes A, B & C

A	B	C
1	1	1
1	1	2
2	1	2
2	2	3
3	3	4

$A \to B$ ✗ $\quad A \to BC$ ✗ $\quad AB \to C$ ✗

$A \to C$ ✗ $\quad B \to AC$ ✗ $\quad \boxed{AC \to B}$ may hold

$B \to A$ ✗ $\quad C \to AB$ ✗

$B \to C$ ✗ $\quad BC \to A$ ✗

$C \to A$ ✗

$\boxed{C \to B}$ may hold based on given relational instance

Find all non-trivial FDs which may hold true in the above relation based on given relational instance.

Ans: $C \to B$ & $AC \to B$

#Q. Write all possible non-trivial FDs with respect to relational schema

R(A, B, C, D)

GATE:- FD $X \to Y$ is called a useful FD if both X & Y are the non-empty sets of attributes and $X \cap Y = \phi$

How many useful FDs are possible in a relation with "4" attributes.

Possible non-trivial FDs over 4 attributes A, B, C & D

A → B
A → C
A → D
B → A
B → C
B → D
C → A
C → B
C → D
D → A
D → B
D → C

A → BC
A → BD
A → CD
B → AC
B → AD
B → CD
C → AB
C → AD
C → BD
D → AB
D → AC
D → BC

A → BCD
B → ACD
C → ABD
D → ABC

28 non-trivial FDs which contain exactly one attribute in LHS.

AB → C
AB → D
AC → B
AC → D
AD → B
AD → C
BC → A
BC → D
BD → A
BD → C
CD → A
CD → B

AB → CD
AC → BD
AD → BC
BC → AD
BD → AC
CD → AB

18 non-trivial FDs which contain exactly two attributes in LHS.

ABC → D
ABD → C
ACD → B
BCD → A

4 non-trivial FDs which contain exactly three attributes in LHS.

∴ Total no. of non-trivial FDs Possible = 28 + 18 + 4 = 50

- Possible non-trivial FDs over 4 attributes A, B, C & D

- Total number of attributes = 4

Case ① When Exactly '1' attribute is present in L.H.S. of $X \longrightarrow Y$

$$^4C_1 * \left(^3C_1 + {}^3C_2 + {}^3C_3\right) = 4*(3+3+1) = 4*7 = 28$$

Choose '1' out of '4' in L.H.S. (and) in R.H.S. we can Choos '1' or '2' or '3' out of 3

Case ② When Exactly 2 attributes are present in L.H.S. of $X \longrightarrow Y$

$$^4C_2 * \left(^2C_1 + {}^2C_2\right) = 6*(2+1) = 6*3 = 18$$

Case ③ When Exactly 3 attributes are present in L.H.S. of $X \longrightarrow Y$

$$^4C_3 * \left(^1C_1\right) = 4*1 = 4$$

$\therefore$ Total no. of FDs = 28 + 18 + 4 = 50

Q: Find the number of non-trivial FDs possible over relation $R(A, B, C, D, E)$

Topic : Properties of Functional Dependencies

① Reflexivity :- If $X \supseteq Y$, then $X \rightarrow Y$ is called a reflexive FD.
Every reflexive FD always holds true in a relation

② Augmentation :- Let functional dependency $X \rightarrow Y$ exist in relation R and 'Z' is some set of attributes over relation R, then $XZ \rightarrow YZ$ will also hold true in relation R

③ Transitivity :- Let FDs $X \rightarrow Y$ and $Y \rightarrow Z$ exists in relation R then $X \rightarrow Z$ will also hold true in relation R

These three properties are called Armstrong's Axiom

④ Decomposition (Splitting Rule) : If $X \rightarrow YZ$ exist in relation R, then $X \rightarrow Y$ as well as $X \rightarrow Z$ will also exist in relation R

If $XY \rightarrow Z$ holds in relation R then $X \rightarrow Z$ and $Y \rightarrow Z$ need not hold in relation R

We are not allowed to split the FDs w.r.t. attributes in L.H.S.

Let, $AB \rightarrow BC$ exists in relation R

$\underbrace{AB \rightarrow BC}_{\text{Semi non-trivial}} \xRightarrow{\text{Apply decomposition}}$ $AB \rightarrow B$ {trivial part}, $AB \rightarrow C$ {Non-trivial part}

We can always decompose a semi non-trivial FD into two parts (1) Trivial Part (useless) & (2) Non-trivial Part (useful)

(5) Union :- If $X \rightarrow Y$ and $X \rightarrow Z$ exists in relation R, then $X \rightarrow YZ$ will also hold true in relation R

(6) Composition: If $X \rightarrow Y$ and $P \rightarrow Q$ exists in relation R, then $XP \rightarrow YQ$ will also hold true in relation R.

(7) Pseudo Transitivity :- If $X \rightarrow Y$ and $YW \rightarrow Z$ exists in rel^n R, then $XW \rightarrow YZ$ will also hold true in rel^n R

Topic : Different types of keys

H.W. Read about :

1. Candidate key
2. Primary key
3. Alternate key
4. Super key

THANK - YOU

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